

Jarvis AI - Project Overview

Jarvis is a hybrid, privacy-first AI OS orchestrator that transforms your Windows PC into an autonomous AI workstation, controllable from any phone browser in the world.

Architecture at a Glance

Component	Technology	Purpose
Inference Server	FastAPI + Uvicorn	Handles all HTTP requests from your phone
Orchestration Graph	LangGraph	Routes commands through an agentic state machine
Local AI	Ollama (Llama 3.2 1B)	Sub-second intent routing and offline fallback
Cloud AI	Gemini (Google AI Studio)	Complex reasoning, vision, web search, code
Text-to-Speech	Piper TTS	On-device English and Hindi voice responses
Secure Tunnel	Ngrok	HTTPS access from anywhere, microphone enabled
Memory	SQLite + ChromaDB	Persistent conversation history and PDF RAG brain

The 3-Tier Speed Router

Tier 1 - Regex (<10ms): Direct OS commands (volume, mute, shortcuts) are matched deterministically and execute instantly.

Tier 2 - FastEmbed/Semantic (<150ms): Fuzzy natural language is matched via ONNX-optimized CPU embeddings. No heavy PyTorch dependencies needed.

Tier 3 - Gemini Cloud (<3s): Complex multi-step tasks, vision, web search, and code generation handled by the cloud AI with a self-healing 3-attempt retry loop.

Security Architecture

Timing-safe PIN: All endpoints use `hmac.compare_digest()` to prevent brute-force timing attacks.

One-time CSRF tokens: Every dangerous action generates a `secrets.token_urlsafe(32)` token that expires after one use.

Human-in-the-Loop Gate: ALL OS-level commands (both local and cloud paths) require explicit phone confirmation before execution.

Prompt Injection Blocker: Hardcoded regex blocks `Invoke-WebRequest`, `iex`, `net user`, and disk format commands even if the AI goes rogue.

Webcam Privacy Beep: An audible beep fires before the webcam activates.

Per-IP Rate Limiting: Maximum 15 requests per minute per device to prevent abuse.

Performance Benchmarks (Phase 4)

Metric	Before	After (Phase 4)
Cold Start Penalty	12 seconds	0s (pinned in RAM)
Simple Command Latency	15 seconds	Under 150ms
Session RAM Leak	+20MB per image	Zero (SQLite off-heap)
API Outage Handling	Server crash	Graceful fallback to local AI
Ngrok Thread Death	Silent failure	Auto-reconnect with backoff